THE CARTESIAN DESCRIPTION OF THE MORPHISM OF BIG AND SMALL ZARISKI SITES

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file topologies.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): the official pool candidate recorded statement [823, 835] and proof(s) [[837, 847]]; this spec covers the complete environment instead (statement 823-835, proof span(s) [(837, 847)]), since in this chapter the author's proof environment(s) follow the statement environment and some pool entries under-counted them.

Editorial note. The computation uses the site-level pushforward formula (Sites construction, a complete cross-chapter reference); it does not rest on the usual-notions identification whose details the author omits elsewhere.

Source reference. Original lemma 020Y: sources/stacks-chow/tags.txt, source lines 2651–2651. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0211: sources/stacks-chow/tags.txt, source lines 2654–2654. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03CF: sources/stacks-chow/tags.txt, source lines 4382–4382. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let Sch_{Zar} be a big Zariski site. Consider a cartesian diagram*

$$\begin{array}{ccc} T' & \xrightarrow{g'} & T \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array}$$

in Sch_{Zar}. Then $i_g^{-1} \circ f_{big,*} = f'_{small,*} \circ (i_{g'})^{-1}$ and $g_{big}^{-1} \circ f_{big,*} = f'_{big,*} \circ (g'_{big})^{-1}$.

Proof. Since the diagram is cartesian, we have for U'/S' that $U' \times_{S'} T' = U' \times_S T$. Hence both $i_g^{-1} \circ f_{big,*}$ and $f'_{small,*} \circ (i_{g'})^{-1}$ send a sheaf $\mathcal{F}$ on $(Sch/T)_{Zar}$ to the sheaf $U' \mapsto \mathcal{F}(U' \times_{S'} T')$ on S'_{Zar} (use Lemmas 020Y and 0211). The second equality can be proved in the same manner or can be deduced from the very general Sites, Lemma 03CF. □

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